



C20-EC-503

7642

BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DECE – FIFTH SEMESTER EXAMINATION

OPTICAL AND MOBILE COMMUNICATIONS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define Snell' Law and cone of acceptance.
2. What are the various fiber optic components?
3. List the sources and detectors used in optical fiber communication.
4. List the limitations of conventional mobile phone system.
5. Define hand off in mobile communication.
6. List the demerits of analog cellular system.
7. List various interfaces used in GSM architecture.
8. List the salient features of GPRS.
9. List the key features of EDGE.
10. List the advantages of 3G over earlier versions.

PART—B

8×5=40

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **eight** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** (a) Define Numerical aperture and derive its expression in terms of refractive indices of core and cladding.

(OR)

- (b) Explain the light wave propagation in optic fiber cable with a diagram.

- 12.** (a) State and explain the working principle of Laser.

(OR)

- (b) Explain the block diagram of DWDM system.

- 13.** (a) Explain the band and Out band signaling systems in Telephony.

(OR)

- (b) Explain the two types of dialing system used in telephony.

- 14.** (a) Explain the Cluster and Capacity of a Cellular system.

(OR)

- (b) Explain the soft and hard hand off systems.

- 15.** (a) Explain the authentication and encryption processes used in GSM security.

(OR)

- (b) Draw and explain the GSM frame structure.

PART—C

10×1=10

- Instructions :**
- (1) Answer the following question.
 - (2) The question carries **ten** marks.
 - (3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

- 16.** Explain the multiple access technic that allows each user to transmit over the entire frequency spectrum for all times.

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